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Praktikum 7

<pre>npx sequelize-cli model:generate --name User --attributes name:string,email:string,password:string,is_active:integer</pre> <pre>npx sequelize-cli db:migrate</pre>	Perintah untuk membuat berkas cetak biru model User beserta migrasinya, lalu mengeksekusinya ke MySQL untuk membentuk tabel users secara otomatis.
--	--

/index.js

mengimpor dan mendaftarkan userRoutes agar semua request dengan prefix path /api/users diteruskan ke router internal user.

```
index.js
const userRoutes = require("../routes/userRoutes");
app.use("/api/users", userRoutes);
```

/routes/userRoutes.js

Mengatur pemetaan (routing) endpoint HTTP method (GET, POST, PUT, DELETE) menuju fungsi aksi spesifik yang ada di dalam userController.

```
userRoutes.js
const express = require("express");
const router = express.Router();
const userController = require("../controllers/userController");

router.get("/", userController.getAllUsers);
router.get("/:id", userController.getUserById);
router.post("/", userController.createUser);
router.put("/:id", userController.updateUser);
router.delete("/:id", userController.deleteUser);

module.exports = router;
```

/models/user.js

Merepresentasikan struktur data tabel objek `User` di aplikasi, lengkap dengan validasi tipe data, nilai default, serta konfigurasi sinkronisasi kolom basis data.

```

user.js

"use strict";
const { Model } = require("sequelize");

module.exports = (sequelize, DataTypes) => {
  class User extends Model {
    static associate(models) {
      //
    }
  }
  User.init(
    {
      id: {
        type: DataTypes.INTEGER,
        primaryKey: true,
        autoIncrement: true,
        allowNull: false,
      },
      name: {
        type: DataTypes.STRING,
        allowNull: false,
      },
      email: {
        type: DataTypes.STRING,
        allowNull: false,
        unique: true,
      },
      password: {
        type: DataTypes.STRING,
        allowNull: false,
      },
      is_active: {
        type: DataTypes.TINYINT,
        allowNull: false,
        defaultValue: 1,
      },
    },
    {
      sequelize,
      modelName: "User",
      tableName: "users",
      underscored: true,
      timestamps: true,
    },
  );
  return User;
};

```

/controllers/userControllers.js

Memuat seluruh logika bisnis utama CRUD (mengambil, menambah, mengubah, menghapus data) dengan memanggil fungsi internal Sequelize ORM untuk memanipulasi data user.

```

userController.js

const db = require("../models");
const User = db.User;

// 1. GET /api/users (Mengambil daftar semua user)
exports.getAllUsers = async (req, res) => {
  try {
    const users = await User.findAll();
    res.status(200).json({
      success: true,
      message: "Berhasil mengambil semua data user",
      data: users,
    });
  } catch (error) {
    res.status(500).json({ success: false, message: error.message });
  }
};

// 2. GET /api/users/:id
exports.getUserById = async (req, res) => {
  try {
    const user = await User.findById(req.params.id);
    if (!user) {
      return res
        .status(404)
        .json({ success: false, message: "User tidak ditemukan" });
    }
    res.status(200).json({
      success: true,
      message: "Berhasil mengambil detail user",
      data: user,
    });
  } catch (error) {
    res.status(500).json({ success: false, message: error.message });
  }
};

// 3. POST /api/users
exports.createUser = async (req, res) => {
  try {
    const { name, email, password, is_active } = req.body;

    // Validasi input
    if (!name || !email || !password) {
      return res
        .status(400)
        .json({
          success: false,
          message: "Name, email, dan password wajib diisi",
        });
    }

    const newUser = await User.create({
      name,
      email,
      password,
      is_active: is_active ?? 1, // Jika tidak diisi, default ke 1 (aktif)
    });

    res.status(201).json({
      success: true,
      message: "User berhasil ditambahkan",
      data: newUser,
    });
  } catch (error) {
    if (error.name === "SequelizeUniqueConstraintError") {
      return res
        .status(400)
        .json({ success: false, message: "Email sudah terdaftar" });
    }
    res.status(500).json({ success: false, message: error.message });
  }
};

exports.updateUser = async (req, res) => {
  try {
    const user = await User.findById(req.params.id);
    if (!user) {
      return res
        .status(404)
        .json({ success: false, message: "User tidak ditemukan" });
    }

    const { name, email, password, is_active } = req.body;

    await user.update({
      name: name || user.name,
      email: email || user.email,
      password: password || user.password,
      is_active: is_active !== undefined ? is_active : user.is_active,
    });

    res.status(200).json({
      success: true,
      message: "Data user berhasil diperbarui",
      data: user,
    });
  } catch (error) {
    if (error.name === "SequelizeUniqueConstraintError") {
      return res
        .status(400)
        .json({
          success: false,
          message: "Email sudah digunakan oleh user lain",
        });
    }
    res.status(500).json({ success: false, message: error.message });
  }
};

exports.deleteUser = async (req, res) => {
  try {
    const user = await User.findById(req.params.id);
    if (!user) {
      return res
        .status(404)
        .json({ success: false, message: "User tidak ditemukan" });
    }

    await user.destroy();
    res.status(200).json({
      success: true,
      message: "User berhasil dihapus",
    });
  } catch (error) {
    res.status(500).json({ success: false, message: error.message });
  }
};

```

/migrations/xxxx-create-user.js

Mengontrol pembentukan fisik skema tabel users di MySQL lengkap dengan detail tipe data, status primary key, indeks unik email, serta timestamp otomatis.

```
20260615032216-create-use

"use strict";
/** @type {import('sequelize-cli').Migration} */
module.exports = {
  async up(queryInterface, Sequelize) {
    await queryInterface.createTable("users", {
      id: {
        allowNull: false,
        autoIncrement: true,
        primaryKey: true,
        type: Sequelize.INTEGER,
      },
      name: {
        type: Sequelize.STRING,
        allowNull: false,
      },
      email: {
        type: Sequelize.STRING,
        allowNull: false,
        unique: true,
      },
      password: {
        type: Sequelize.STRING,
        allowNull: false,
      },
      is_active: {
        type: Sequelize.TINYINT,
        allowNull: false,
        defaultValue: 1,
      },
      created_at: {
        allowNull: false,
        type: Sequelize.DATE,
        defaultValue: Sequelize.literal("CURRENT_TIMESTAMP"),
      },
      updated_at: {
        allowNull: false,
        type: Sequelize.DATE,
        defaultValue: Sequelize.literal(
          "CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP",
        ),
      },
    });
  },
  async down(queryInterface, Sequelize) {
    await queryInterface.dropTable("users");
  },
};
```

TEST API

ronWW's Workspace

Overview GET View Book POST Create U PUT Update Boo GET View All U GET View User PUT Update U DEL delete us No environment

Users > Create Users

POST http://localhost:3001/api/users

Docs Params Authorization Headers 8 Body Scripts Settings Cookies

none form-data x-www-form-urlencoded raw binary GraphQL JSON

```
1 {
2   "name": "Violet Evergarden",
3   "email": "Violet@example.com",
4   "password": "password123",
5   "is_active": 1
6 }
```

Body Cookies Headers 10 Test Results

201 Created 8 ms 590 B Save Response

{ } JSON Preview Visualize

```
2  "success": true,
3  "message": "User berhasil ditambahkan",
4  "data": {
5    "id": 5,
6    "name": "Violet Evergarden",
7    "email": "Violet@example.com",
8    "password": "password123",
9    "is_active": 1,
10   "updatedAt": "2026-06-15T03:56:10.702Z",
11   "createdAt": "2026-06-15T03:56:10.702Z"
12 }
13 }
```

Full-screen Snip

Connect Git Console Terminal 0 0

Globals Vault Tools

ronWW's Workspace

Overview GET View Book POST Create U PUT Update Boo GET View All U GET View User PUT Update U DEL delete us No environment

Users > View All Users

GET http://localhost:3001/api/users

Docs Params Authorization Headers 6 Body Scripts Settings Cookies

Body Cookies Headers 10 Test Results

200 OK 12 ms 929 B Save Response

{ } JSON Preview Visualize

```
3  "message": "Berhasil mengambil semua data user",
4  "data": [
5    {
6      "id": 1,
7      "name": "Imron",
8      "email": "imron@example.com",
9      "password": "password123",
10     "is_active": 1,
11     "createdAt": "2026-06-15T03:48:09.000Z",
12     "updatedAt": "2026-06-15T03:48:09.000Z"
13   },
14   {
15     "id": 2,
16     "name": "Xandrea",
17     "email": "Xandrea@example.com",
18     "password": "password123",
19     "is_active": 1,
20     "createdAt": "2026-06-15T03:53:10.000Z",
21     "updatedAt": "2026-06-15T03:53:10.000Z"
22   },
23   {
24     "id": 3,
25     "name": "TEST USER",
```

Connect Git Console Terminal 0 0

Globals Vault Tools

ronWW's Workspace

Overview GET View Book POST Create U PUT Update Boo GET View All U GET View User PUT Update U: DEL delete us No environment

Items + ...

COLLECTIONS

- Books
 - POST Create Books
 - GET View Book by ID
 - PUT Update Book
 - DEL DELETE BOOK
 - GET View All Book
- My Collection
- Users
 - GET Create Users
 - GET View All Users
 - GET View User by ID
 - GET Update User
 - GET delete user

ENVIRONMENTS

SPECS

FLOWS

Users > View User by ID

GET http://localhost:3001/api/users/5

Send

Docs Params Authorization Headers 6 Body Scripts Settings Cookies

Query Params

Key	Value	Description	Bulk Edit
Kev	Value	Description	

Body Cookies Headers 10 Test Results

{ } JSON Preview Visualize

```
1 {
2   "success": true,
3   "message": "Berhasil mengambil detail user",
4   "data": {
5     "id": 5,
6     "name": "Violet Evergarden",
7     "email": "Violet@example.com",
8     "password": "password123",
9     "is_active": 1,
10    "createdAt": "2026-06-15T03:56:10.000Z",
11    "updatedAt": "2026-06-15T03:56:10.000Z"
12  }
13 }
```

200 OK • 11 ms • 590 B • Save Response

ronWW's Workspace

Overview GET View Book POST Create U PUT Update Boo GET View All U GET View User PUT Update U: DEL delete us No environment

Items + ...

COLLECTIONS

- Books
 - POST Create Books
 - GET View Book by ID
 - PUT Update Book
 - DEL DELETE BOOK
 - GET View All Book
- My Collection
- Users
 - GET Create Users
 - GET View All Users
 - GET View User by ID
 - GET Update User
 - GET delete user

ENVIRONMENTS

SPECS

FLOWS

Users > Update User

PUT http://localhost:3001/api/users/3

Send

Docs Params Authorization Headers 8 Body Scripts Settings Cookies

none form-data x-www-form-urlencoded raw binary GraphQL JSON

Schema Beautify

```
1 {
2   "name": "User Diperbarui",
3   "email": "USER.new@example.com",
4   "is_active": 0
5 }
```

Body Cookies Headers 10 Test Results

{ } JSON Preview Visualize

```
1 {
2   "success": true,
3   "message": "Data user berhasil diperbarui",
4   "data": {
5     "id": 3,
6     "name": "User Diperbarui",
7     "email": "USER.new@example.com",
8     "password": "password123",
9     "is_active": 0,
10    "createdAt": "2026-06-15T03:53:35.000Z",
11    "updatedAt": "2026-06-15T04:00:31.740Z"
12  }
13 }
```

200 OK • 15 ms • 589 B • Save Response

ronWW's Workspace

Search

Invite Upgrade

OverviewGET View BookPOST Create UPUT Update BoolGET View All UGET View UserPUT Update U:DEL delete us

Users > delete user

SaveShare

DELETE

http://localhost:3001/api/users/2

Send

DocsParamsAuthorizationHeaders6BodyScriptsSettings

Query Params

Key	Value	Description	Bulk Edit	...
Key	Value	Description		

BodyCookiesHeaders10Test Results

200 OK • 12 ms • 391 B • Save Response

JSON

PreviewVisualize

```
1 {
2   "success": true,
3   "message": "User berhasil dihapus"
4 }
```

ENVIRONMENTS

SPECS

FLows

Connect GitConsoleTerminal00

GlobalsVaultTools

11:04 AM